Progress Report: OPDC-Aligned Validation Pipeline

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1. Context & Objective

This work tests whether data-driven PD subtypes replicate across independent cohorts - PPMI ($n = 794$, drug-naïve, OFF-state) and OPDC Discovery ($n = 996$, treated, ON-state) - using GAMLSS normalisation and bootstrap consensus clustering [1, 3]. The OPDC-aligned experimental phase extends the baseline $k = 2$ replication to 9 clinical features covering 7 domains and evaluates imaging validation using the SWI and DaTscan data provided by Tamir and Johannes.

2. Baseline Cross-Cohort Replication ($k = 2$, 5 features)

The initial analysis (Table 2) compared four harmonisation strategies. CORAL was selected for its near-maximal cross-cohort ARI combined with the lowest symmetry gap, ensuring balanced bidirectional transfer (PPMI→OPDC ARI = 0.723; OPDC→PPMI ARI = 0.693).

Harmonisation	ARI (500-boot)	Silhouette	Sym. gap	
None	0.142	0.284	0.322	
Pooled-Z	0.142	0.284	0.322	
ComBat-EB	0.797	0.327	0.156	
CORAL	**0.756**	**0.324**	**0.010**	selected

3. Five-Subtype Discovery (PPMI, $n = 821$, 9 features)

The expanded pipeline uses 9 GAMLSS-normalised features: MDS-UPDRS I (non-motor), II (ADL), III (motor exam), MoCA (cognition), Schwab & England (function), RBDSQ (REM-sleep behaviour), ESS (sleepiness), orthostatic SBP drop (dysautonomia), and STAI/HADS-A (anxiety). A 144-configuration grid search ($K \times cov \times imp \times harm$) yielded 5 clinically distinct subtypes (Table 3).

Label	n (%)	Dominant features	Clinical profile
S1	104 (12.7)	UPDRS-III 19.7, MoCA 27.6, SE 94.4	Moderate motor, preserved cognition and function
S2	222 (27.0)	UPDRS-III 14.7, SE 100, RBD 3.1	**Mildest:** low motor severity, full ADL independence, minimal RBD
S3	121 (14.7)	UPDRS-III 29.9, SE 77.6, MoCA 25.8	**Most severe:** high motor + ADL burden, cognitive decline, elevated RBD
S4	278 (33.9)	UPDRS-III 23.1, SE 90.0, RBD 4.2	Intermediate motor-predominant, moderate functional impact
S5	96 (11.7)	UPDRS-I 3.1, ESS 8.0, SBP drop 5.5	Non-motor predominant: sleepiness, dysautonomia, preserved function

4. Robust Findings from the Extended Validation

The following results meet criteria of statistical significance, clinical coherence, and (where applicable) cross-cohort replication:

- **Longitudinal Kaplan-Meier (PPMI):** All 5 clinical milestones (UPDRS-III > 27, MoCA < 25, ESS > 8, SCOPA-AUT > 14, RBDSQ > 6) show significant separation by subtype (log-rank $p < 0.0001$, all 5 tests). S3 progresses fastest; S2 slowest.
- **SuStaIn staging (PPMI):** Across all 4 reconstructed progression sequences, UPDRS-III is con-

sistently the last biomarker to reach abnormality - consistent with ascending (non-motor→motor) disease propagation.

- **OPDC cross-cohort ARI ($K = 5$):** Mean bidirectional ARI = 0.254 (matched OPDC $n = 135$). Lower than $k = 2$, as expected for finer-grained subtyping.
- **SWI nigrosome-1 (iPD only, $n = 53$):** After PPMI-profile matching, only ∼10 iPD patients overlap with the subtyped sub-sample, distributed across 3 of 5 subtypes (S1: $n = 1$, S2: $n = 4$, S5: $n = 5$). Kruskal-Wallis tests non-significant ($p > 0.3$, $q_{BH} = 1.0$). Reported as an honest power limitation.

5. Imaging Data - Current Status

- **SWI (June 2021 snapshot):** 53 iPD; 30 have follow-up scan (flag = 1) - an **unexploited opportunity** for longitudinal within-subject nigrosome-1 analysis.
- **DaTscan SUR (May 2019):** 9 iPD with quantitative caudate/putamen ratios; 8 overlap with SWI. Insufficient alone for subtype-level inference.
- **Gap:** Any post-2021 SWI timepoints for iPD (per Griffanti et al. [2]) and genetics/PRS would extend the validation substantially.

6. Proposed Next Steps

1. Exploit 30 iPD follow-up SWI scans for longitudinal nigrosome-1 change analysis, stratified by baseline subtype.
2. Investigate whether additional SWI/DaTscan timepoints for iPD exist beyond the June 2021 dataset [2].
3. Complete the extended validation remotely from UMONS (2 months); Prof. Benjelloun available for data-sharing agreements.

Key Highlights for Publication

Bimodal replication confirmed: ARI = 0.756, symmetry gap = 0.010 (PPMI $n = 794$ ↔ OPDC $n = 996$) [1].
5-subtype Kaplan-Meier: all 5 milestones significant ($p < 0.0001$); S3 (diffuse-severe) vs S2 (mild) define the prognostic extremes.
SuStaIn: motor exam (UPDRS-III) is universally the last event in the progression cascade - supporting non-motor-first disease models.

References

[1] Lawton M et al. Developing and validating PD subtypes and their motor and cognitive progression. *JNNP* 2018;89(12):1279-87.

[2] Griffanti L et al. OPDC Discovery longitudinal MRI substudy. *BMJ Open* 2020;10:e034110.

[3] Marras C, Hu MT et al. MDS Task Force: purpose-driven PD subtyping. *Mov Disord* 2024.

[4] Hani M et al. PPMI vs OPDC cohort comparison and harmonisation strategy. Working document v3, March 2026.